



Junhao Wei



澳門理工大學
Universidade Politécnica de Macau
Macao Polytechnic University

Birthday: April 2001 **e-mail:** p2312195@mpu.edu.mo

Affiliation: Macao Polytechnic University

Research Interests: Robotics, Trajectory Generation, RL, VLA

Educational Background

2023.08-2025.06	Macao Polytechnic University	Big Data & Internet of Things
-----------------	------------------------------	-------------------------------

GPA: 3.2

Main courses: Machine Learning, Big Data Analysis, Convolutional Neural Networks, Internet of Things Communication, Fundamentals of the Internet of Things, AIDD

2019.09-2024.06	Beijing Institute of Technology, Zhuhai	UAV System Engineering
-----------------	---	------------------------

GPA: 3.2

Main courses: Signals and Systems, Digital Signal Processing, Introduction to Intelligent Control Technology, Principles of Flight, Aerodynamics, High-Frequency Electronic Circuits, Fundamentals of Mechanical Design, Engineering Mechanics, Analog Circuits, Digital Circuits.

Projects

UAV Path Planning Based on Improved Particle Swarm Optimization (PSO)

- **Project Topic:** To address the issue of the Particle Swarm Optimization (PSO) algorithm getting trapped in local optima, an optimization approach is proposed that uses Tent mapping initialization + dynamic parameter adjustment + adaptive t-distribution + adaptive position update.
- **Adaptive Position Update Particle Swarm Optimization for UAV Path Planning**
(Core b WIOPT2024: Accept)
- **Independently Completed · First Author**

RNav: A Safe Trajectory Planner based on Reinforcement Learning in dynamic environments

- **Project Topic:** RNav is an end-to-end path planning framework for mobile robots, based on deep reinforcement learning (DRL) using the Proximal Policy Optimization (PPO) algorithm. Unlike traditional planning methods that decompose decision-making into separate modules, RNav directly learns a robust navigation policy from experience. This eliminates the need for handcrafted models and manual parameter tuning. RNav's key strength lies in its ability to navigate both static and dynamic environments, adapting to moving obstacles. The system uses carefully designed state and action representations to enable the robot to make safe decisions in real-time. A unique safety shield, inspired by velocity obstacles, is incorporated to prevent failures due to the black-box nature of neural networks. For training, RNav employs parallel simulations with multiple robots in NVIDIA Isaac Sim, accelerating convergence and scalability. Experimental results show that RNav outperforms traditional methods, ensuring safe and efficient navigation with fewer collisions. Its end-to-end learning approach makes it a powerful solution for autonomous navigation in dynamic environments.

An Aerial Manipulation System with Visual Autonomous Grasping and Master-Slave Control

- **Project Topic:** This project develops an aerial manipulation UAV combining a robotic arm, D435 depth camera, and Orin NX onboard computing. With servo-based attitude feedback and perception-driven control, the aerial platform supports autonomous object grasping, teleoperation, and fully autonomous navigation in cluttered environments.

XLeRobot Dual-Arm Mobile Manipulation

- **Project Topic:** To enable practical household manipulation tasks, a low-cost dual-arm mobile robot is built based on LeRobot framework,integrating SO101 arms with Lekiwi mobile base, and leveraging VLA(Vision-Language-Action) models, VR teleoperation (XLeVR), and imitation learning for general-purpose robotic control.

6D Robotic Arm Grasping

- **Project Topic:** To pick objects from a chaotic pile, PnP pose solving, ICP, and deep GraspNet are used to control the pose and position, and LLM and VLM are called to process text and image data.

An Enhanced WOA with Multi-strategy

- **Project Topic:** To address the shortcomings of the WOA , such as low convergence accuracy, slow convergence, and the tendency to get trapped in local optima, this paper proposes an enhanced Whale Optimization Algorithm (LSEWOA). The LSEWOA uses Good Nodes Set Initialization and employs newly designed Leader-Followers Search-for-Prey Strategy, Spiral-based Encircling Prey Strategy, and Enhanced Spiral Updating Strategy for updating positions. It redesigns the update mechanism of the coefficient "a," balancing the global exploration ability and local exploitation ability of the WOA. Its performance significantly outperforms other SOTA metaheuristic algorithms.
- *LSEWOA: An Enhanced Whale Optimization Algorithm with Multi-strategy for Numerical And Engineering Design Optimization Problems*
(JCR Q2 Sensors: Accept)
- Independently Completed · First Author

Credit Card Fraud Detection Based on Improved XGBoost

- **Project Topic:** An improved SVM SMOTE method is proposed, combined with MiniBatchKMeans, to address the high dimensionality, large data volume, and extreme class imbalance in credit card transaction data. This approach further enhances the effectiveness and accuracy of XGBoost in credit card fraud detection, giving it stronger generalization and anti-interference capabilities.
- *Credit Card Fraud Detection Based on MiniKM-SVM SMOTE-XGBoost Model*
(BDIOT2024 Accept)
- Cooperation Completed · Second Author

FPCA: Dynamic Feature and Clinical Correlation Analysis

- **Project Topic:** This study proposed the FPCA framework to enhance patient representation learning from EHR by addressing two issues: (1) underutilization of dynamic feature patterns, and (2) neglected feature correlations across clinical phases. The framework includes the FCF and MFE modules to improve clinical prediction accuracy.
- **Dual-perspective Patient Representation Learning based on Electronic Health Records**
(JCR Q1 Neurocomputing: Submitted)
- Cooperation Completed · Fourth Author

Internship Experience

2024.06-2024.12	Pazhou Lab (Huangpu)	Algorithm Intern
• GAC Aion APS System Optimization • Responsible for optimization work in the Guangdong-Hong Kong-Macao Greater Bay Area (Huangpu) International Algorithm Competition. • Participated in the 12345 dispatch project, responsible for the deployment and fine-tuning of large models.		

Campus Experience

2024.09 - now	Macao Polytechnic University	Research Assistant
• The UAV research group, affiliated with Faculty of Applied Sciences at the Macao Polytechnic University; • Primarily responsible for UAV path planning ; • Completing the projects assigned by the supervisor.		
2020.09 - 2023.06	Beijing Institute of Technology, Zhuhai	Vice Class Leader
• Organizing and managing class activities, as well as handling activity reimbursements; • Developing functional plans for each department and coordinating their development; • Assisting the counselor in student communication tasks.		
2019.09 - 2020.06	Beijing Institute of Technology, Zhuhai	Learning Monitor
• Managing various academic tasks for the class • Assisting the teacher with tasks between the teacher and students		

Awards

- 2025 Macao Polytechnic University PhD Full Scholarship
- The 20th "Huawei Cup" China Graduate Mathematical Modeling Competition - "Third Prize"
- The 18th "Challenge Cup" National University Students Extracurricular Academic Science and Technology Works Competition - "Second Prize"
- Aviation College, 2020-2021 Academic Year - "Outstanding Student Leader"
- Aviation College, 2020-2021 Academic Year, First Semester - "First-class Scholarship"
- Aviation College, 2020-2021 Academic Year, Second Semester - "Second-class Scholarship"
- Aviation College, 2019-2020 Academic Year, Second Semester - "Second-class Scholarship"
- Aviation College, 2021-2022 Academic Year, First Semester - "Third-class Scholarship"
- Beijing Institute of Technology (Zhuhai), 2022-2023 Academic Year - "Team Second Prize" in Badminton Competition
- Beijing Institute of Technology (Zhuhai), 2024 First "School Anniversary Cup" Bay Area Alumni Badminton Tournament - Runner-up

Skills

- CET-6, CET-4, able to communicate fluently with foreigners
- Familiar with C and Python, knowledgeable in Linux
- Proficient in CAD, Catia, Matlab, Altium Designer, Keil, QGC, and other professional software; familiar with simulation platforms such as Simulink, ROS, and Gazebo
- Fluent in Cantonese

Publication (First author)

- ***LSEWOA: An Enhanced Whale Optimization Algorithm with Multi-strategy for Numerical And Engineering Design Optimization Problems***
Junhao Wei, Yanzhao Gu, Yuzheng Yan, Zikun Li, Baili Lu, Shirou Pan and Ngai Cheong*
JCR Q2 Sensors: **Accept (Featured Paper)**
- ***RWOA: A Novel Enhanced Whale Optimization Algorithm with Multi-strategy for Numerical Optimization And Engineering Design Problems***
Junhao Wei, Yanzhao Gu, Baili Lu, Ngai Cheong*
JCR Q1 PLOS ONE: **Accept**
- ***MRBMO: An Enhanced RBMO Algorithm for Solving Numerical Optimization Challenges***
Junhao Wei, Yanzhao Gu, Yuzheng Yan, Zikun Li, Baili Lu, Shirou Pan, Ngai Cheong*, Ruishen Zhou
JCR Q2 Symmetry: **Accept**
- ***LSWOA: An Enhanced Whale Optimization Algorithm with Levy flight and Spiral flight for Numerical And Engineering Design Optimization Problems***
Junhao Wei, Yanzhao Gu, Zhanxi Xie, Yuzheng Yan, Yapeng Wang, Baili Lu, Zikun Li, Ngai Cheong*, Jiafeng Zhang, Song Zhang
JCR Q1 PLOS ONE: **Accept**
- ***Adaptive Position Updating Particle Swarm Optimization algorithm for UAV path planning***
Junhao Wei, Yanzhao Gu, Eddie Law*, Ngai Cheong
Core b 2024 22nd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks: **Accept**
- ***TSWOA: An Enhanced WOA with Triangular Walk and Spiral Flight for Engineering Design Optimization***
Junhao Wei, Yanzhao Gu, Yuzheng Yan, Yapeng Wang, Zikun Li, Baili Lu, Shirou Pan, Ngai Cheong*
IEEE 8th International Conference on Advanced Algorithms and Control Engineering (ICAACE2025) : **Accept**
- ***ESTGWOA: a novel enhanced WOA with multi-strategy for solving engineering design optimization challenges***
Junhao Wei, Yanzhao Gu, Zikun Li, Baili Lu, Shirou Pan, Yapeng Wang, Ngai Cheong*
JCR Q1 Scientific Reports: **Accept**
- ***Modified Rapidly-exploring Random Tree Algorithm for Urban UAV Pathing Planning***
Junhao Wei, Zhanxi Xie, Jiafeng Zhang
IEEE 11th International Conference on Mechanical Engineering, Materials and Automation Technology: **Accept**
- ***AHRRT: An Enhanced Rapidly-Exploring Random Tree Algorithm with Heuristic Search for UAV Urban Path Planning***
Junhao Wei, Yanzhao Gu, Ran Zhang, Wenxuan Zhu, Shuai Wu, Yapeng Wang , Ngai Cheong, Zhiwen Wang, Sio-Kei Im, Xu Yang*
JCR Q2 Symmetry: **Submitted**
- ***CICDWOA: A Collective Cognitive Sharing Whale Optimization Algorithm with Cauchy Inverse Cumulative Distribution for 2D/3D Path Planning and Various Engineering Applications***
Junhao Wei, Ran Zhang, Wenxuan Zhu, Yanzhao Gu, Yanxiao Li, Yifu Zhao, Zikun Li, Ngai Cheong, Zhiwen Wang, Sio-Kei Im, Yapeng Wang, Xu Yang*
JCR Q1 / SCI Q1 top Knowledge-based Systems: **Submitted**

Publication (First author)

- **GeoSSA: Geometric Sparrow Search Algorithm for UAV Path Planning and Engineering Design**

Junhao Wei, Wenxuan Zhu, Ran Zhang, Yanzhao Gu, Yanxiao Li, Yifu Zhao, Zikun Li, Ngai Cheong, Zhiwen Wang, Sio-Kei Im, Yapeng Wang, Xu Yang*

JCR Q1 Cluster Computing: **Submitted**

- **AHMWOA: An Geometric WOA for Numerical Optimization and Engineering Design Optimization Challenges**

Junhao Wei, Ran Zhang, Wenxuan Zhu, Yanzhao Gu, Yanxiao Li, Yifu Zhao, Zikun Li, Ngai Cheong, Zhiwen Wang, Sio-Kei Im, Yapeng Wang, Xu Yang*

JCR Q1 / SCI Q2 top Applied Soft Computing: **Submitted**

- **RBMOOBL: An Enhanced RBMO Algorithm for Solving Numerical Optimization Challenges**

Junhao Wei, Wenxuan Zhu, Ran Zhang, Yanzhao Gu, Yanxiao Li, Yifu Zhao, Zikun Li, Ngai Cheong, Zhiwen Wang, Sio-Kei Im, Yapeng Wang, Xu Yang*

JCR Q1 Scientific Reports: **Submitted**

- **An Enhanced Whale Optimization Algorithm with Log-Normal Distribution for Optimizing Coverage of Wireless Sensor Networks**

Junhao Wei, Yanzhao Gu, Ran Zhang, Wenxuan Zhu, Yapeng Wang, Xu Yang and Ngai Cheong*

The 5th International Conference on Artificial Intelligence, Information Processing and Cloud Computing : **Accept**

Publication (Non-first author)

- **GWOA: A Multi-Strategy Enhanced Whale Optimization Algorithm For Engineering Design Optimization**

Yanzhao Gu, **Junhao Wei**, Zikun Li, Baili Lu, Shirou Pan, Ngai Cheong*

JCR Q1 PLOS ONE: **Accept**

- **Credit Card Fraud Detection Based on MiniKM-SVMSMOTE-XGBoost Model**

Yanzhao Gu, **Junhao Wei**, Ngai Cheong*

the 2024 8th International Conference on Big Data and Internet of Things: **Accept**

- **WSLWOA: An Enhanced Whale Optimization Algorithm with Cauchy-Levy Mutation for Solving Numerical and Engineering Design Optimization Problems**

Yanxiao Li, **Junhao Wei**, Yanzhao Gu, Ran Zhang, Dexing Yao, Haochen Li, Xu Yang, Sio-Kei Im, and Yapeng Wang*

JCR Q1 Cluster Computing: **Under Review**

- **STWOA-XGBoost: A Novel Hybrid Approach for Robust Early-Stage Prediction of Academic Potential in Computer Science Students**

Yanzhao Gu, **Junhao Wei**, Ngai Cheong*

JCR Q1 Cluster Computing: **Under Review**

- **Dual-perspective Patient Representation Learning based on Electronic Health Records**

Yuzheng Yan, Ziyue Yu, KaKit Chio, **Junhao Wei**, Hongxu Yuan and Wuman Luo*

JCR Q1 Neurocomputing: **Under Review**

Patent

- **Fully Automatic Intelligent Chinese Medicine Decoction Equipment**

Authors: Ngai Cheong, Ning Zhao, Muya Yao, Yufei Hu, Yanzhao Gu, **Junhao Wei**

Status: **Application Submitted**

Master's Graduation Project

- **AHRRT: an enhanced Rapidly-exploring Random Tree algorithm with heuristic search for UAV urban path planning**
- **Project Topic:** This paper considers that the paths generated by the RRT algorithm have poor quality and slow planning speed. It proposes an enhanced RRT algorithm (AHRRT). ERRT integrates probabilistic target bias strategy, adaptive step size strategy, improved path pruning optimization strategy, and B-spline smoothing strategy. In the experiments, we compare AHRRT with common path planners (A* algorithm, ACO algorithm, RRT* algorithm, traditional RRT algorithm). The results show that, in various scenarios, ERRT demonstrates strong planning capabilities and can efficiently generate higher-quality paths.

Self-Evaluation

I have an outgoing, steady, and energetic personality. I am warm-hearted, sincere, and responsible in my work, with a proactive attitude. I am willing to endure hardships and enjoy thinking critically. I am open-minded in communication with others. I possess strong organizational skills, practical hands-on abilities, and team collaboration spirit. I can quickly adapt to various environments and integrate into them. I have a strong sense of social responsibility, am diligent and hardworking, and actively seek opportunities for personal growth. I continuously strive to improve myself and enhance my overall qualities. In addition to excelling in my major and general courses, I have also read a large number of extracurricular books, constantly gaining new knowledge and broadening my horizons. I actively participate in social practice activities and various cultural and sports events, which have helped me develop a certain level of organizational and social skills.